🧩 Assignment 08: Working with Dictionaries in Python

📘 Instructions

1. Write each program in a separate Python file.
2. Use meaningful variable names and proper indentation.
3. Display clear output messages.
4. Use dictionary operations wherever applicable.
5. Test your programs with different values.

📖 1️⃣ Create and Display a Dictionary

Problem:

Create a dictionary to store information about a student such as name, age, and grade. Display the complete dictionary.

Example Output

```
Student = {'name': 'Ahmed', 'age': 16, 'grade': 'A'}
```

🔍 2️⃣ Access Dictionary Values

Problem:

Create a dictionary containing information about a book and display the title and author separately.

Example Output

```
Book Title: Python Basics
Author: John Smith
```

➕ 3️⃣ Add a New Key-Value Pair

Problem:

Create a dictionary containing employee information. Add a new key called `salary` and display the updated dictionary.

Example Output

```
Original Dictionary = {'name': 'Ali', 'department': 'IT'}

Updated Dictionary = {'name': 'Ali', 'department': 'IT', 'salary': 50000}
```

✏️ 4️⃣ Update a Dictionary Value

Problem:

Create a dictionary containing product details. Update the price of the product and display the updated dictionary.

Example Output

```
Original Dictionary = {'product': 'Laptop', 'price': 45000}

Updated Dictionary = {'product': 'Laptop', 'price': 50000}
```

❌ 5️⃣ Remove an Item from a Dictionary

Problem:

Create a dictionary of student details and remove one key-value pair using the `pop()` method.

Example Output

```
Original Dictionary = {'name': 'Sara', 'age': 15, 'grade': 'B'}

Updated Dictionary = {'name': 'Sara', 'grade': 'B'}
```

📋 6️⃣ Display All Keys and Values

Problem:

Create a dictionary of countries and capitals. Display all keys and all values separately.

Example Output

```
Countries:
India
Japan
France
```

```
Capitals:
New Delhi
Tokyo
Paris
```

🔄 7️⃣ Iterate Through a Dictionary

Problem:

Create a dictionary of subjects and marks. Use a loop to display each subject and its corresponding mark.

Example Output

```
Math : 85
Science : 90
English : 78
```

🧮 8️⃣ Calculate Total Marks from a Dictionary

Problem:

Given a dictionary containing subject names and marks, calculate the total marks obtained by the student.

Example Output

```
Marks = {'Math': 85, 'Science': 90, 'English': 78}

Total Marks = 253
```

⭐ Bonus Question

Student Report Card

Problem:

Create a dictionary containing a student's name and marks in three subjects. Display:

- Student Name
- Marks in each subject
- Total Marks
- Average Marks

Example Output

```
Student Name: Ahmed

Math: 80
Science: 90
English: 85

Total Marks = 255
Average Marks = 85.0
```

🎯 Learning Outcomes

After completing this assignment, students will be able to:

✅ Create and use dictionaries ✅ Access values using keys ✅ Add, update, and remove dictionary items ✅ Traverse dictionaries using loops ✅ Work with dictionary keys and values ✅ Perform calculations using dictionary data ✅ Build simple real-world applications using dictionaries